

Dataset, compute, and training configuration

Dataset

Compute and training

Dataset	v2 (Isaac Sim 5.1, Go2 USD)	Hardware	2× NVIDIA RTX 5060 Ti (16 GB each)
Trajectories (clean)	1,008 (FALLs excluded)	Software	PyTorch 2.11 + CUDA 13.0 (Blackwell sm_120)
Total frames	3,624,696 (~33 hrs at 30 Hz)	Loss	MSE on body-frame velocity, weighted [1,1,1,5]
Frame rate	30 Hz	Optimiser	AdamW, lr 1e-4, weight decay 1e-4
Cameras	stereo, 1280 × 720 RGB + RAFT-Stereo disparity	LR schedule	ReduceLROnPlateau (factor 0.5, patience 3)
Sensors	IMU 9-axis (accel, gyro, mag), 12-DOF joints	Modality dropout	10 % per-sample per-modality (fusion only)
Terrain (clean)	forest 633 · flat 288 · uneven 126 · uphill 14	Clip length / stride	fusion: 40 / 8 · Stage-2: 8 / 8
Train / val / test split	650 / 150 / 208 (stratified by terrain, seed 42)	Batch size	fusion: 128 · Stage-2 (CNN): 16 · MViT: 4
Test clips (fusion)	92,935 (clip_len=40, stride=8)	Patience	10 epochs
Test clips (Stage-2)	93,767 (clip_len=8, stride=8)	Preprocessing wall-clock	kin 5 min · ins 8 min · vo (DROID) ~2.4 days
Label	6-D body-frame velocity [vx vy vz ωx ωy ωz]	Best fusion training time	fusion_v2: ~82 minutes